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2029

Week 2 problem 5 Approximate continuous function with a smooth one.

Some exam questions from hws

There'll probably be a partitions of unity question.

Week 3 Problem 2. Show the torus is a manifold.

There will probably be a regular value theorem in the question.

"prove this thing is a manifold" easiest way is with regular value theorem.

$$4 \quad F: \mathbb{R}^n \rightarrow \mathbb{R}^n$$

$$f(x_1, \dots, x_n) = \left(x_1^{\frac{1}{3}}, x_2, \dots, x_n\right) \quad (1)$$

Claim

$$F(M) = F(g^{-1}0) = (g \circ f^{-1})^{-1}0 \quad (2)$$

just an exercise in set theory

Proof we'll show 0 is a regular value of g^{-1} .

Theorem If $h: \mathbb{R}^n \rightarrow \mathbb{R}$ is smooth such that 0 is a regular value.

missing something XXX

0 is a regular value of